

Real In-cell Integrated Touch Sensor for Flexible OLED Display

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Abstract

This work presents a novel structure for integrating capacitive touch sensor within thin film encapsulation (TFE) of OLED display. Through patterning the cathode with an undercut isolation structure, one region is allocated for display and another for touch sensing. The proposed design ensures normal display while overcoming the critical challenge of high capacitive load in the in-cell structure. Finally, a 1.43-inch prototype was developed, which exhibits sensitive touch and superior display quality. The adoption of the in-cell structure not only eliminates the need for the external touch sensor manufacturing process outside TFE but also reduces the overall module thickness.

Author Keywords

capacitive touch sensor; integration; in-cell; OLED; undercut isolation

1. Introduction

In the field of LCD displays, touch technology has undergone the evolution from add-on, to on-cell, and finally to in-cell. The increasing integration of touch structures with LCD displays not only enables a thinner display module but also enhances its optical transmittance and brightness.

For OLED displays, the technological path for the touch structure is also transitioning from the add-on to on-cell. Particularly in product such as wearables, smartphones, and foldable devices, the on-cell touch structure has become the mainstream solution. This structural evolution has led to a significantly thinner module, enhanced panel flexibility, improved light transmittance, and a reduced risk of optical mura [1] (Figure 1a and 1b). Progressing from on-cell to in-cell integration allows for a thinner module

thickness, and thus lowers the risk of TP layer fracture during OLED bending processes. Through integrating the touch structure within the thin-film encapsulation (TFE) layer, the obstruction of TP electrodes to the light emitted by pixels is decreased, leading to an enhanced optical performance. In addition, the adoption of the in-cell touch structure eliminates the need for the TP fabrication process after TFE, which reduces the number of the required photomasks and optimizes the manufacturing cost of OLED display modules (Figure 1c).

However, the realization of the in-cell touch structure in AMOLED displays presents two major challenges. First, the OLED display features a common cathode that is distributed across the entire surface. When the touch structure is integrated into the TFT array, the electric field generated by the touch sensor are shielded by the common cathode, preventing the detection of finger touch. Second, the in-cell touch structure significantly reduces the distance between the touch electrodes and the cathode, anode, and other signal lines. This not only leads to a large capacitive load, but also causes interference between the display and touch functions.

In this paper, we propose a novel design that utilizes cathode patterning technology to segment the cathode into two independent parts: one serving as the display cathode and the other as the touch electrode. As illustrated in Figure 2, the cathode above the undercut structure acts as the touch electrode, while the cathode in the pixel aperture area remains used for the OLED display. This strategy not only resolves the cathode shielding issue but also eliminates the manufacturing process for touch electrodes on the TFE encapsulation layer, offering advantages in cost reduction, module thinning, and enhanced light transmittance.

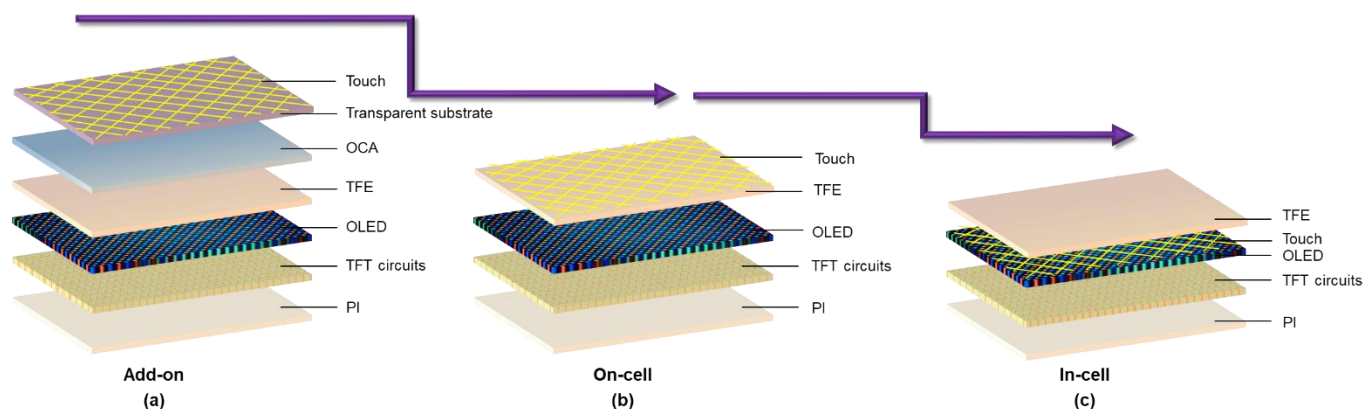


Figure 1. The comparison of the stacked structures for different OLED display modules: (a) Add-on touch; (b) On-cell touch; and (c) In-cell touch.

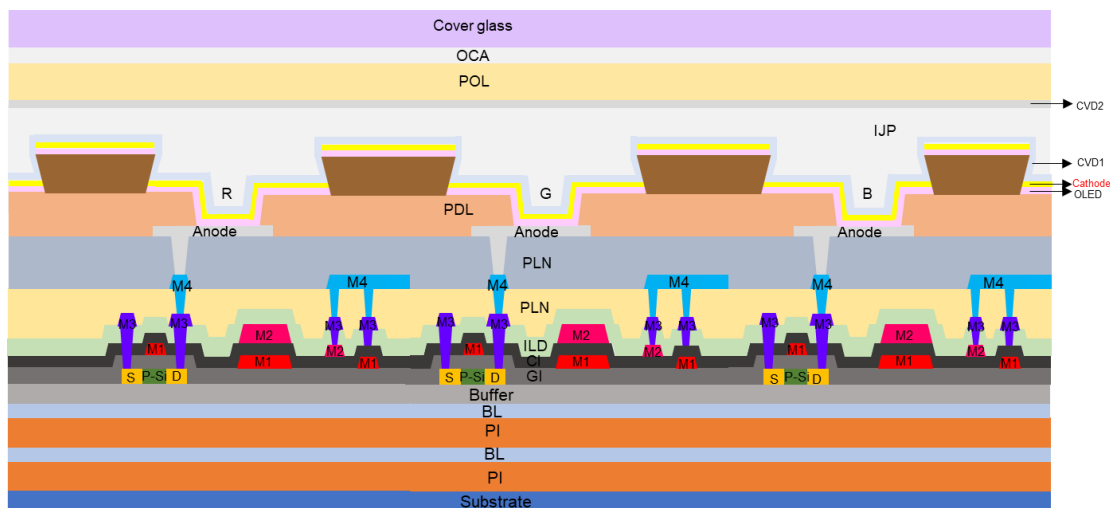


Figure 2. Schematic illustrating the stacked structure of an in-cell touch integrated OLED display module.

2. Design and Experiments

2.1 Sensor Design

To address the issue of electric field shielding by the common cathode, the cathode must be patterned. The main methods for cathode patterning include laser etching [2], utilizing cathode patterning materials (CPM) [3], and adopting isolation structures. Laser etching faces the risk of damaging the OLED pixels during cathode ablation, and involves high equipment costs, while utilizing CPM materials are limitations in patterning precision and resolution.

This work presents a scheme that utilizes the undercut isolation structures for cathode patterning, defining one region as the OLED cathode and the other one as the touch electrode (the cathode on the isolation structures), as shown in Figure 2. In addition, isolations are placed within the gaps between pixels to form a touch electrode grid, simultaneously guaranteeing the in-plane connectivity of the display cathode (Figure 3).

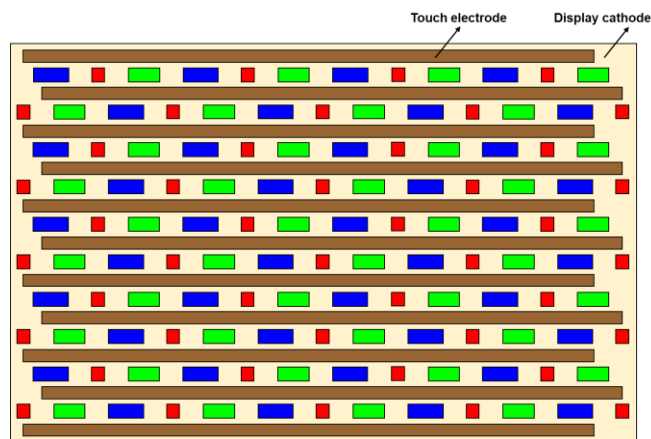


Figure 3. Schematic diagram showing the distribution of the touch electrode and display cathode.

2.2 Sensor Fabrication

To achieve reliable and complete cathode isolation, it is necessary to fabricate isolation structures with a reverse-tapered profile.

Figure 4 illustrates the specific preparation process. The process began with coating the negative photoresist after the pixel definition layer (PDL) was fabricated. Subsequently, the negative photoresist was patterned into undercut isolation structures through a series of processing steps: soft baking, exposure, post-exposure baking (PEB), development, and hard baking. Figure 5 shows the morphology of the resulting structure, with corresponding taper angle, membrane thickness, and line width of approximately 78.9° , $1.26 \mu\text{m}$, and $11.36 \mu\text{m}$, respectively. After OLED organic functional materials and cathode evaporation onto this undercut structure, complete isolation can be achieved. The fabrication for in-cell integrated touch sensor requires only one additional coating and photolithography steps in the array process, along with one extra photomask, offering significant advantages in both process feasibility and cost control.

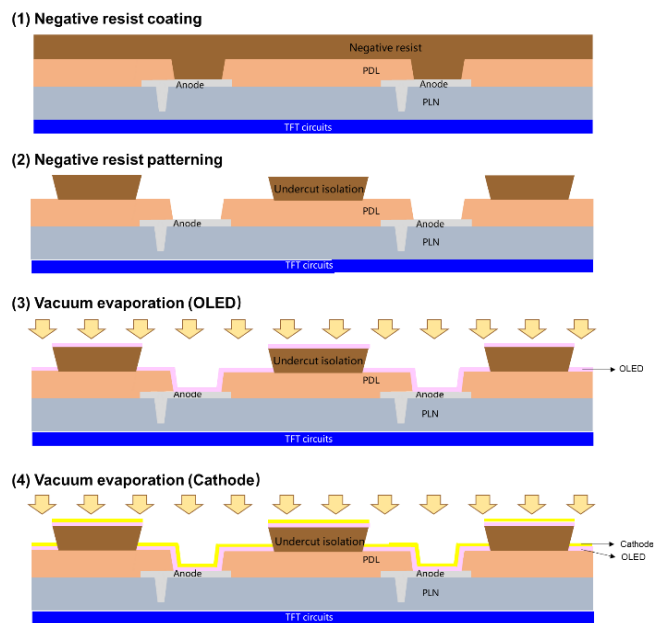


Figure 4. Schematic diagram of the fabrication process.

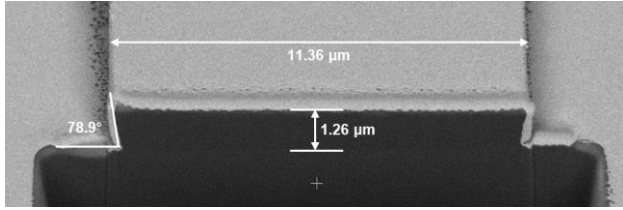


Figure 5. Morphology of the 1.2 μm -thick undercut isolation.

2.3 Performance Improvement

In in-cell integrated structure design, the spacing between the touch electrodes and the cathode, anode, and other signal lines is significantly reduced. This not only results in a high capacitive load, but also easily causes mutual interference between the touch and display functions. To address these issues, the solution adopted in this work is to increase the membrane thickness of the undercut structure.

The simulation results demonstrate that increasing the thickness of undercut structure from 1.2 μm to 2.5 μm leads to a reduction in the base capacitance (C_p) by 22.25% and an improvement in signal strength (ΔC) by 40.91%, as detailed in Table 1. In the in-cell touch structure employed in this work, the capacitive coupling between the TP electrode and both the cathode and the anode is predominantly lateral, with the parallel-plate capacitance being negligible. Moreover, due to the extremely small spacing between the TP electrode and the cathode/anode, the lateral electric field strength is significantly greater than that in on-cell touch structure (Figure 6). Through increasing the thickness of the undercut isolation to increase the distance between the TP electrode and the cathode/anode, the lateral capacitive coupling is suppressed. This strategy enables more electric field lines from the TP electrode to project outward and couple with the finger, rather than being confined by the internal electrodes, which enhances touch sensitivity.

Table 1. Simulation results for different undercut structures with thicknesses of 1.2 μm and 2.5 μm .

Item	1.2 μm	2.5 μm
C_p (pF)	121.8	94.7
C_p' (pF)	122.02	95.01
ΔC (pF)	0.22	0.31
$\Delta C/C_p$	0.18%	0.33%

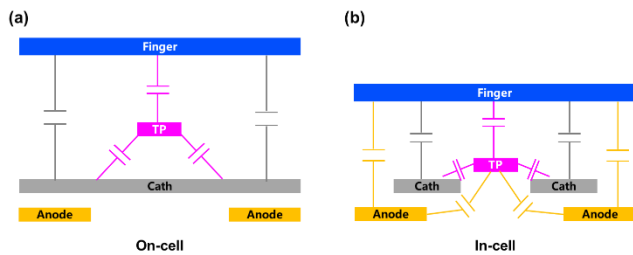


Figure 6. Schematic diagram showing the capacitive coupling in different touch sensor configurations: (a) On-cell touch; (b) In-cell touch.

3. Prototype

A prototype of a 1.43-inch OLED display module with an in-cell integrated touch sensor has been developed, with its main display parameters summarized in Table 3. The in-cell panel in the display state exhibits any visible Mura, as presented in Figure 7. The touch performance was evaluated using a grounding copper pillar with a diameter of 7 mm to simulate the human finger. Figure 7 confirms that the 7 mm copper pillar is capable of scribing sensitively on the sample surface. Benefitting from the high signal and low noise level, its statistical SNR is calculated to be higher than 35dB. More detailed touch test results are provided in Table 3. In addition, the display performance was also assessed. The results indicate that the display quality meets the requirements for daily use, and the corresponding measurement results are summarized in Table 4.

Table 2. Specifications of the prototype.

Item	Specifications
Panel Size	1.43inch
Resolution	466RGB*466
Substrate	Polyimide
Encapsulation	TFE
Process	LTPS
Pixel Density	326ppi
Touch Type	Self-capacitance
Touch Channels	32

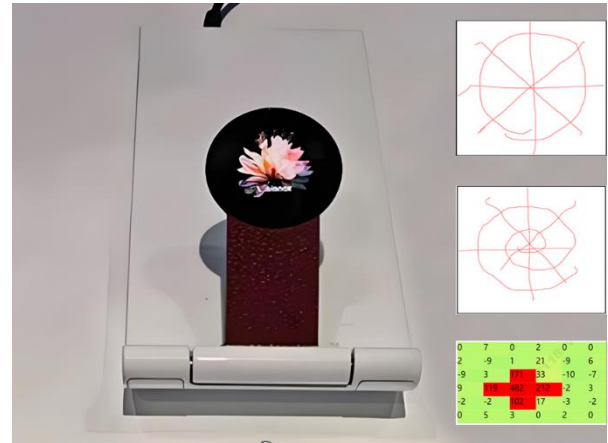


Figure 7. Prototype of the 1.43-inch OLED display module with an in-cell integrated touch sensor.

Table 3. Test results for touch performance.

Touch Performance	Spec.	Test Value
Report Rate	120Hz	120Hz
Display White SNR	>35dB	37dB
Response Time	<30ms	22ms
Two-Finger Separation	<15mm	11mm
Accuracy & Linearity ($\Phi 7$ mm)	Center <1.0mm Edge area <1.5mm	Center <1.0mm Edge area <1.5mm
Jitter ($\Phi 7$ mm)	Center <0.4mm Edge area <0.5mm	Center <0.4mm Edge area <0.5mm

Table 4. Test results for display performance.

Display Performance	Spec.	Test Value
Brightness	600nits	620nits
Boost Mode Bright	1200nits	1260nits
Time Color Temperature	7500±500K	7431K
CR	1000000:1	3137500
Uniformity	90%	91%
Flicker	-30dB	-47dB
Cross Talk	<1.1%	1.04%
DCI-P3(CIE1931)	100%±3%	102%

4. Conclusion

In this work, we designed and fabricated the undercut isolation structure to divide the cathode into two parts: one for the OLED cathode, and the other one for the touch electrode. This strategy addresses the cathode shielding issue, allowing for the realization of a real in-cell integrated touch sensor. By patterning the touch electrodes into multiple independent blocks that are uniformly distributed across the entire display area, an in-plane conductive network is successfully achieved among the display cathodes of all

pixels. Finally, a 1.43-inch circular smartwatch with an in-cell touch structure was developed. The touch measurements demonstrate that the prototype exhibits a touch SNR greater than 35 dB, a reporting rate of 120 Hz, and high sensitivity for smooth drawing. Display performance tests show that key metrics such as brightness, contrast ratio, and color gamut are adequate for normal usage scenarios.

5. References

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